

Day_2_Descriptive Statistics

September 8, 2026

1 Day 2 - Descriptive Statistics

Descriptive Statistics are used to summarize and understand numerical data using mean , meadin, standard deviation , quartiles, standard correlation.

```
[2]: import pandas as pd
df = pd.read_excel(
    "Day_4_Ensemble_Learning_Customer_Churn.xlsx",
    sheet_name="Customer Churn Data"
)
df
```

```
[2]:
```

	CustomerID	Age	Tenure_Months	MonthlyCharges	Contract	\
0	1001	25	3	65	Month-to-month	
1	1002	34	18	72	One year	
2	1003	45	36	85	Two year	
3	1004	29	7	55	Month-to-month	
4	1005	52	48	95	Two year	
5	1006	41	24	78	One year	
6	1007	23	2	50	Month-to-month	
7	1008	38	30	88	One year	
8	1009	49	42	92	Two year	
9	1010	31	12	60	Month-to-month	
10	1011	27	8	68	Month-to-month	
11	1012	56	60	105	Two year	
12	1013	44	28	82	One year	
13	1014	36	20	75	One year	
14	1015	30	15	58	Month-to-month	
15	1016	47	40	90	Two year	
16	1017	22	4	45	Month-to-month	
17	1018	39	32	80	One year	
18	1019	53	54	100	Two year	
19	1020	33	16	70	Month-to-month	

	InternetService	TechSupport	PaymentMethod	Churn
0	Fiber optic	No	Electronic check	Yes
1	DSL	Yes	Bank transfer	No
2	Fiber optic	Yes	Credit card	No

3	DSL	No	Electronic check	Yes
4	Fiber optic	Yes	Credit card	No
5	DSL	Yes	Bank transfer	No
6	No	No	Mailed check	Yes
7	Fiber optic	Yes	Electronic check	No
8	Fiber optic	Yes	Credit card	No
9	DSL	No	Mailed check	Yes
10	Fiber optic	No	Electronic check	Yes
11	No	Yes	Bank transfer	No
12	DSL	Yes	Credit card	No
13	Fiber optic	No	Electronic check	Yes
14	DSL	No	Mailed check	Yes
15	Fiber optic	Yes	Credit card	No
16	No	No	Mailed check	Yes
17	DSL	Yes	Bank transfer	No
18	Fiber optic	Yes	Credit card	No
19	DSL	No	Electronic check	Yes

1.1 1. Mean

Mean represents the average value of a numerical column.

```
[3]: df.mean(numeric_only=True)
```

```
[3]: CustomerID      1010.50
Age                37.70
Tenure_Months      24.95
MonthlyCharges     75.65
dtype: float64
```

1.2 2. Median

Median represents the middle value of the numerical data.

```
[4]: df.median(numeric_only=True)
```

```
[4]: CustomerID      1010.5
Age                37.0
Tenure_Months      22.0
MonthlyCharges     76.5
dtype: float64
```

1.3 3. Standard Deviation

Standard deviation shows how much the data values vary from the mean.

```
[9]: df.std(numeric_only=True)
```

```
[9]: CustomerID      5.916080
      Age           10.387746
      Tenure_Months  17.479236
      MonthlyCharges 16.865646
      dtype: float64
```

1.4 4. Quartiles

Quartiles divide numerical data into four parts. The 25%, 50%, and 75% values represent Q1, Q2, and Q3.

```
[12]: df.describe()
```

```
[12]:      CustomerID      Age  Tenure_Months  MonthlyCharges
count      20.00000  20.000000      20.000000      20.000000
mean     1010.50000  37.700000      24.950000      75.650000
std         5.91608  10.387746      17.479236      16.865646
min      1001.00000  22.000000       2.000000      45.000000
25%      1005.75000  29.750000      11.000000      63.750000
50%      1010.50000  37.000000      22.000000      76.500000
75%      1015.25000  45.500000      37.000000      88.500000
max      1020.00000  56.000000      60.000000     105.000000
```

1.5 5. Correlation

Correlation measures the strength and direction of the linear relationship between numerical variables.

```
[13]: df.corr(numeric_only=True)
```

```
[13]:      CustomerID      Age  Tenure_Months  MonthlyCharges
CustomerID      1.000000  0.119044      0.192644      0.085189
Age              0.119044  1.000000      0.979092      0.951385
Tenure_Months    0.192644  0.979092      1.000000      0.948489
MonthlyCharges    0.085189  0.951385      0.948489      1.000000
```

1.6 Conclusion

Descriptive statistics help summarize the Customer Churn dataset. Mean and median show the central values, standard deviation shows data variation, quartiles describe the distribution, and correlation helps identify relationships between numerical variables.